THE MATH CIRCLE

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1. **Week 1: Introduction**

19 Oct 2018

The goal of the first week is introducing the whole idea of the Math Circle, and as a gift (!), proving the nice theorem of Brouwer. From the next week, the circle will officially start its course with Mathematical Induction.

1.1. **Sperner's Lemma.** Let ABC be a triangle. Suppose we have triangulated ABC (by possibly adding vertices inside or on the edges of ABC). Color the vertices as follows:

- Color A, B, and C with 1, 2 and 3, respectively.
- Color the vertices on the edges of ABC with one of the colors of the endpoints, e.g. color the vertices lying on AB with either 1 or 2.
- Color the rest of the vertices arbitrarily.

Prove that one can always find a simple triangle (i.e., a triangle without any vertex inside it) having all three colors on its vertices.

Note 1.1. There are several ways to attack this problem. Induction, if used correctly, can be one way. However, there is an elegant proof just using parity. A sketch of the argument is as follows:

> Let T denote the set of simple triangles in the triangulation of ABC. Build a graph G as follows: the vertices of G correspond to triangles in T. There is an extra vertex in G which corresponds to the area outside of ABC. We connect two vertices of G with an edge, if and only if their corresponding areas share a common segment with two endpoints colored with 1 and 2. Now just notice that a graph can only have an even number of vertices having an odd degree.

Notice that this argument proves a stronger result: there is an odd number of simple triangles with all three colors.

1.2. **Sperner's Lemma for Higher Dimensions.** Can you state and prove a similar argument for dimensions higher than 2?

1.3. **Gift: Brouwer's Fixed Point Theorem.** Let D be the unit closed disk in $\mathbb{R}^2$ and $f : D \to D$ be a continuous function. Prove that f has a fixed point, i.e. there exists some $x \in D$ with $f(x) = x$.

Note 1.2. The Brouwer's Fixed Point Theorem is more general than what we proved. Its complete statement is as follows:

> Let f be a continuous function mapping a compact[1] convex set to iteself. Then f has a fixed point.

[1] A set $C \subseteq \mathbb{R}^n$ is *compact* if and only if it is bounded and closed. By closedness we mean that C includes all of its limit points.

2. **Week 2: Induction (I)**

26 Oct 2018

2.1. **Reachability.** Let a, b, c, and d be four positive real numbers. Starting with the tuple (a, b, c, d), we make a new tuple (ab, bc, cd, da), and continue as follows: we multiply each element by its next one, and multiply the last element by the first element. Prove that by repeating this procedure, we will never reach (a, b, c, d) again, unless $a = b = c = d = 1$.

2.2. **Sparse Matrix.** Let $a_1, \ldots, a_m$ and $b_1, \ldots, b_n$ be positive integers, such that

$$a_1 + \cdots + a_m = b_1 + \cdots + b_n.$$

Prove that one can put at most $m + n - 1$ positive numbers in an $m \times n$ matrix, such that for all $i \in [m]$ and $j \in [n]$, sum of the elements in ith row would be equal to a_i, and sum of the elements in jth column would be equal to b_j.

2.3. **Walking Bee.** We have tiled the plane using regular hexagons with side length 1. A bee, walking on the edges of the network, moves from point A to point B in 100 moves (which we know is the shortest path). Prove that for half of his journey, he has moved in one direction (out of 3 possible directions).

2.4. **Missing Planet.** On each planet of a planetary system, an astronomer is observing the nearest planet to himself. We know that the distances between the planets are different numbers. Prove that if the number of planets is odd, then there is a planet which is unobserved. We assume that the planets have 0 radii.

2.5. **Gift: Antipodal Points on Sphere.** Draw a closed curve on a sphere in $\mathbb{R}^3$, dividing it into two equal parts. By equality, we mean that one part can be rotated and fitted on the other part. Examples for such divisions are dividing the sphere into two hemispheres by the equator, or the tennis ball's division made by the S-curve on the ball.

Prove that one can always find two antipodal points on the curve, i.e., two points such that they are collinear with the center of the sphere.

3. **Week 3: Permutations (I)**

2 Nov 2018

This week is about *structural* properties of permutations. Here I bring the most important notes:

- A permutation is a bijection $\pi : I \to I$ for some set I. Usually I is a countable set, and in our case, is the finite set $[n] = \{1, \dots, n\}$, for some natural number $n \in \mathbb{N}$.
- One can construct a digraph from π: the vertices are $[n]$ and i is connected to $\pi(i)$. This graph is a disjoint union of cycles.
- One can represent each permutation in terms of its cycles in the following way: Sort the cycles by their smallest element. Starting from 1, write down the cycle in clock-wise order ending in 1, and proceed to the next cycles.
- In problems involving *partial information* about a permutation (like the Gift below), the following function turns out to be useful:
$$\bar{\pi}(i) = \#\{1 \le j \le i : \pi(j) \ge \pi(i)\}.$$
One can prove that for a random permutation π, the random variables $\bar{\pi}(i)$ are all independent.

3.1. **Cycles.** We choose a permutation π uniformly at random from S_n, the set of all permutations of $[n]$. Compute

(1) the probability that π has no fixed points.
(2) the probability that π has just one cycle.
(3) the probability that the cycle containing 1 has k elements.
(4) elements 1 and 2 belong to the same cycle.
(5) the expected number of cycles.

3.2. **Jewellery business.** We have n saving boxes, with different keys. Someone locks the boxes and throws the keys into them at random, putting one key in each box. We break open k boxes. What is the probability that we are now able to open all the rest?

3.3. **Records.** What is the expected number of on-going records at a long jump competition, in which each of the n competitors has one jump, these jumps are all different and the starting order is drawn at random?

3.4. **Gift: Betting Strategy.** Imagine a long jump competition, in which no two of the n competitors jump the same distance. They jump in a drawn order. One can bet according to the following rule: After a jump, one can say "This is best of all." We arrive on the scene after kth jump. What is the probability p_k that, following the best strategy, we will guess the winner correctly (information about previous results is available)? What strategy should we follow in order to have the highest probability of winning (provided we arrive on time)?

4. **Week 4: League (I)**

9 Nov 2018

4.1. **Polygons in a Regular Polygon.** How many convex k-gons can be formed from the vertices of a regular n-gon, if two k-gons are considered essentially different when they do not arise from each other by rotation?

4.2. **Colorations.** What is the number of k-colorations of the vertices of a regular n-gon, if two k-colorations are considered as essentially different when they arise by rotation from each other?

4.3. **Scales and Students.** On the desk there is a scale. On the scalepans there are some weights (possibly of different grams). On the weights, family names of one or more students are written. As a student comes inside the class, she puts the weights with her name to the other scalepan. Prove that the professor can let certain people to come to the class, such that the heavier scalepan becomes the lighter scalepan afterwards.

4.4. **Uni Language.** In a university, there are 50 students fluent in English, 50 fluent in French, and 50 fluent in Persian (there could be multilingual people). Prove that one can divide the students into 5 groups (not necessarily of equal size), such that in each group there are 10 students fluent in English, 10 fluent in French, and 10 fluent in Persian.

4.5. **General Remarks and Hints.**
Problem 4.1. One can start by computing the number of k-gons that are invariant by a given rotation. Another thing to compute is to compute the number of rotations that keep a k-gon fixed. The final answer is

$$\frac{1}{n}\sum_{d|(n,k)}\binom{n/d}{k/d}\varphi(d),$$

where $\varphi(\cdot)$ is the Euler's phi function.
Problem 4.2. I figured out that a similar argument to Problem 4.1 results in a solution! The answer is

$$\frac{1}{n}\sum_{d|n}\varphi(d)\,k^{n/d}.$$

Problem 4.3. Konstantinos provided a solution using induction. There is also another argument without induction, which I explained more-or-less in the class, but will elaborate more in the next session.
Problem 4.4. Andisheh provided a nice argument, using induction on the number of the groups. I also have an argument, which basically creates a group that every language is spoken by 2 people, using case-by-case analysis.

5. **Week 5: Permutations (II)**

16 Nov 2018

Note 5.1. This week is the last week we talk about permutations. From next week we continue on Induction.

5.1. **Dancing Rings.**

(1) Let $x_1, \ldots, x_n$ be real numbers. For each permutation π of $[n]$ define

$$a(\pi) = \max\{0, x_{\pi(1)}, x_{\pi(1)+\pi(2)}, \ldots, x_{\pi(1)} + \cdots + x_{\pi(n)}\}.$$

Also consider the cycles $C_1, \ldots, C_k$ of π and set

$$b(\pi) = \sum_{l=1}^{k} \max\Big\{0, \sum_{j \in C_l} x_j\Big\}.$$

Then prove that $\{a(\pi) : \pi \in S_n\}$ and $\{b(\pi) : \pi \in S_n\}$ are equal as collections.

(2) m boys and m girls form rings for a dance at random (there may be pairs or even singles). Prove that the probability that in each ring the number of boys is equal to the number of the girls is $\frac{1}{m+1}$.

5.2. **Burnside's Lemma.** Let G be a group of permutations of $[n]$ with k orbits. Then the average number of fixed points of permutations in G is k.
Hint. Maybe starting to think about the following would make the problem more accessible:

$$|\mathrm{Orbit}(x)| \cdot |\mathrm{Stab}(x)| = |G|, \quad \forall x \in X.$$

5.3. **Weighted Version.** Let G be a permutation group acting on a set X. Suppose that a weight $w(x)$ is assigned to each $x \in X$ such that $w(x)$ is invariant under G, i.e., for any given orbit Θ of G, $w(x)$ takes the same value $w(\Theta)$ for all $x \in \Theta$. Prove that

$$\sum_{\Theta} w(\Theta) = \frac{1}{G} \sum_{\pi \in G} \sum_{\pi(x)=x} w(x).$$

5.4. **Gift: Irrationalities in Ford-Fulkerson.** Provide an example where Ford-Fulkerson algorithm for solving the Max-Flow problem does not terminate, involving irrational numbers. (The capacities should be rational, though).

6. **Week 6: Duality and Parity (I)**

23 Nov 2018

6.1. **Potentials.** We associate a value $v(e)$ with each edge of a digraph G. This may be considered as the work needed to go from the tail of e to the head of e. To go conversely, we need $-v(e)$ work. We want to find a potential, i.e., a function $p(x)$ defined on $V(G)$ such that if $e = (x, y)$ then $v(e) = p(y) - p(x)$. Show that this is possible if and only if going around any circuit, the total work is 0.

6.2. **De Bruijn.** Show that we can associate 0 or 1 with each point of a 2^k-cycle ($k > 2$) in such a way that all arcs of length k of the cycle give different 01-sequences.

6.3. **Labyrinth Problem.** Starting at a point x_0 we walk along the edges of a connected graph G according to the following rules:

- We never use the same edge twice in the same direction.
- Whenever we arrive at a point $x \neq x_0$ not previously visited, we mark the edge along which we entered x. We use the marked edge to leave only if we must, i.e. if we have used all the other edges before.

Show that we get stuck at x_0 and that, by then, every edge has been traversed in both directions.

6.4. **Lights and Buttons.**

(a) The set of points of any graph G can be partitioned into two classes V_1, V_2 such that both V_1 and V_2 span subgraphs with even degrees.

(b) For any graph G, $V(G)$ can be partitioned into two classes V_1, V_2 such that V_1 spans a subgraph with even degrees and V_2 spans a subgraph with odd degrees.

(c) Assume that at each vertex of a graph there is a light and a button. At the beginning, all lights are on. Pushing a button will change the status of the light at the vertex and at its neighbors. Show that one can push some buttons so that all lights will be turned off.

7. **Week 7: Parity and Duality (II)**

30 Nov 2018